Experiment 2 – Optoelectronic energy conversion

Explanation: goal, principle, procedure and why we need it later

1. The experiment in one sentence

A solar cell is a photodiode in the 4th quadrant: we measure its dark and bright characteristics, determine I_{SC}, U_{OC}, MPP, fill factor and efficiency, the band gap from the temperature dependence, and calculate the annual yield and electricity cost.

2. Goal of the experiment

- Measure the spectrum of the LED sun simulator, compare it with AM1.5G; determine the integrated irradiance.
- Dark curves at several temperatures: I_S, n, R_S, R_P, band gap W_G.
- Bright curves at 25–100 % intensity: I_{SC}, U_{OC}, MPP, FF, η and η(intensity).
- Cells with bypass LEDs and a fan: observe the effect of intensity.
- Estimate the annual yield and cost per kWh (Duisburg, Tarifa).

3. Principle – the logic behind it

3.1 From the pn junction to the solar cell

Photons with $E \geq W_G$ create electron-hole pairs; the field of the space-charge zone separates them and produces the photocurrent $I_{SC} \propto$ irradiance. The dark curve is the diode curve $I_D = I_S(e^{U/(nU_T)} - 1)$; illuminated: $\mathbf{I = I_{SC} - I_D}$ (equivalent circuit: current source ‖ diode ‖ R_P, in series R_S). $I_S \propto \exp(-W_G/(nkT))$ rises very fast with temperature – this gives the band gap (ln I_S versus 1/T) and the fall of U_{OC} with T (≈ −2 mV/K).

3.2 Characteristic numbers

- $\mathbf{I_{SC}}$ (U = 0) = $\eta_{ext}(e/h\nu)P_{opt}$ – linear in intensity.
- $\mathbf{U_{OC}}$ (I = 0) = $nU_T \ln(I_{SC}/I_S + 1)$ – logarithmic in intensity.
- **MPP:** maximum of P = U·I; $\mathbf{FF = P_{MPP}/(U_{OC}\, I_{SC})}$; $\mathbf{\eta = P_{MPP}/P_{opt} = FF \cdot U_{OC} \cdot I_{SC}/P_{opt}}$.
- $\mathbf{R_S}$ **↑:** curve flatter at U_{OC}, FF ↓. $\mathbf{R_P}$ **↓:** slope at U = 0, U_{OC} ↓ at low light.

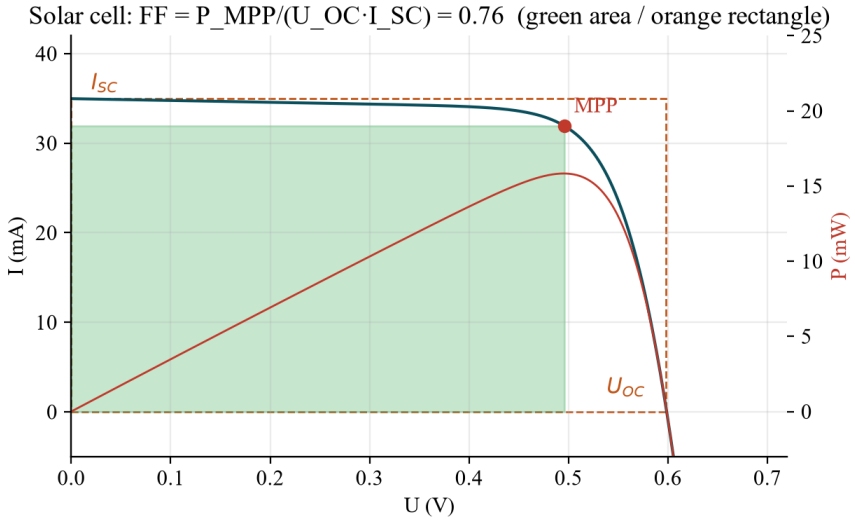


Fig. 1: Bright curve with I_{SC}, U_{OC}, MPP and fill factor (model).

3.3 Intensity, temperature, losses

At C-fold intensity $I_{SC} \rightarrow C \cdot I_{SC}$ and $U_{OC} \rightarrow U_{OC} + nU_T \ln C$ (≈ +70 mV per decade); the efficiency therefore rises slowly with intensity ($\eta(C)/\eta(1) \approx 1 + nU_T \ln C/U_{OC}$) and falls at very high intensity through I^2R_S losses and heating. **Losses:** reflection, non-absorption ($E < W_G$), thermalisation ($E > W_G$), recombination, R_S/R_P. This gives the optimum band gap ≈ 1.1–1.5 eV (Si 1.12 eV). The LED spectrum of the simulator has almost no infrared (spectral mismatch to AM1.5G).

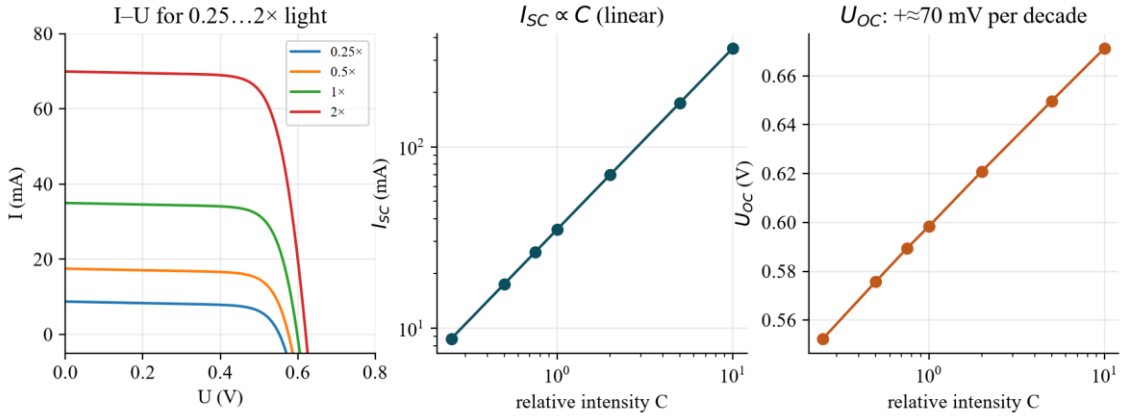


Fig. 2: Intensity dependence (model).

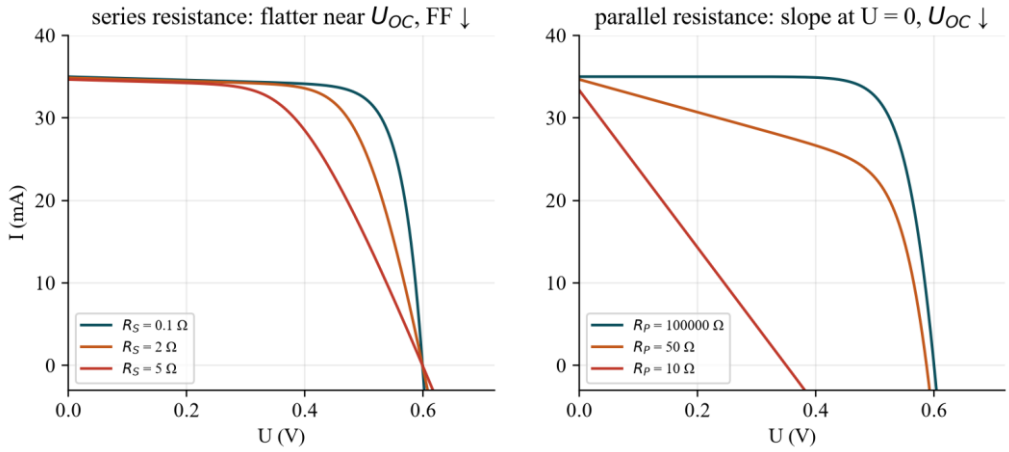


Fig. 3: Effect of R_S and R_P.

3.4 Annual yield and cost

$E_{year} = P_{year} \cdot \eta$ (constant at 1000 W/m²) or $\Sigma\, P_{year} \cdot a_i \cdot \eta(S_i)$ (irradiance classes). Duisburg ≈ 1000 kWh/m²/a, Tarifa 1919 kWh/m²/a. Example η = 15 %: 1 kWp = 6.67 m² → Duisburg ≈ 1000 kWh/kWp/a → at 4000 €/kWp and 20 years ≈ 0.20 €/kWh; Tarifa ≈ 0.10 €/kWh.

4. Procedure step by step

Task	What to do?	Evaluation
(a)	Spectrometer (getSpec-3648, 360–1100 nm): spectrum of the simulator in µW/(cm²·nm)	$E = \int E_\lambda d\lambda$; compare with AM1.5G
(b)	Measure the cell area	$P_{opt} = E \cdot A$
(c)	Dark curves at ≥ 4 temperatures (Keithley B2901A)	I_S, n, R_S, R_P; ln $I_S(1/T) \rightarrow W_G$
(d)	Bright curves at 25/50/75/100 %	I_{SC}, U_{OC}, MPP, FF, η; η(intensity)
(e)	Cell ∥ LED (antiparallel), two strings in series with the fan; intensity 25–100 %	Table: fan speed, LED brightness

Evaluation (§6 I–VII) and Excel template: *06_Auswertung/Auswertung_Versuch2.xlsx*. Optional LTspice demos: Solarzelle_IU_Intensitaet.asc, Solarzelle_IU_Serienwiderstand.asc.

5. What to watch in the lab

- Plan (c) and (d) **with the supervisor** so that W_G and η(intensity) can be determined reliably.
- The cell heats up under light – let it cool after each run, note the temperature.
- Note the area (with/without contact fingers) and the simulator setting (PWM) for every curve.
- Polarity of the bypass LED (antiparallel to the cell!).
- Save the data as a U, I table (USB stick) – the Excel template calculates everything.

6. Why do we need this later?

- **Experiment 1:** the photodiode in the 4th quadrant is the solar cell; short-circuit current ∝ light.
- **Experiment 4:** LED and solar cell are "the same device in the opposite direction"; the band gap governs both.
- **Experiments 3/5/6:** the same diode equation, the same temperature dependence (I_S, U shift).
- Practice: photovoltaics, energy transition, sensing, space; economics = efficiency × location.

7. Typical questions – short answers

Question	Short answer
How does a solar cell work?	$E \geq W_G$ → pairs; the field of the space-charge zone separates them → photocurrent.
I-V equation?	$I = I_{SC} - I_S(e^{U/(nU_T)} - 1)$ (with R_S, R_P).
FF and η?	$FF = P_{MPP}/(U_{OC}\, I_{SC})$; $\eta = P_{MPP}/P_{opt}$.
Intensity dependence?	I_{SC} linear; U_{OC} logarithmic; η rises slowly.
Temperature dependence?	I_S ↑ exponentially → U_{OC} ↓ (≈ −2 mV/K), η ↓.
Determine the band gap?	ln I_S versus 1/T: slope = $-W_G/(nk)$.
R_S and R_P?	R_S ↑: FF ↓; R_P ↓: leakage, U_{OC} ↓.
Loss mechanisms?	Reflection, non-absorption, thermalisation, recombination, ohmic losses.
Optimum band gap?	≈ 1.1–1.5 eV: compromise between photon number and voltage.
AM1.5G / STC?	Reference spectrum, 1000 W/m², 25 °C.
Bypass diode?	Carries the string current around a shaded cell.
Annual yield and cost per kWh?	$E = P_{year} \cdot \eta$; cost = 4000 €/kWp ÷ (20 a · kWh/kWp/a).

Date: **3 Feb 2027**, pre-test (Antestat) 14:00, building LT, room LT 227 (group 9).